

StockTwits: Comprehensive records of a financial social media platform from 2008 to 2022

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This paper describes and characterizes the relationship between social media and financial markets, focusing on two key questions: (1) Do users of financial social media perform better than guessing at predicting stock movements? (2) When do users in the aggregate perform better than guessing at the same task? To study these questions, we introduce the first publicly available, comprehensive dataset of posts on a social media platform: StockTwits. StockTwits is a financial social media platform where more than 7 million active users have discussed financial markets and investing strategies across over 550 million posts since 2008. We provide a complete record of all StockTwits posts up to 2022, including the poster’s anonymous ID, the text and timestamp of the message, and whether the user tagged their post as optimistic (“bullish”) or pessimistic (“bearish”). We study the temporal dynamics of this dataset, analyzing it at both the ticker level and the user level. First, we find that while most users perform approximately at a guessing rate, a significant percentage consistently perform statistically better *or worse* than guessing, especially at longer timescales. Second, corroborating existing literature, we observe that *attention*—as measured by the number of messages

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mentioning a stock—is generally less predictive than *sentiment*. However, there is meaningful variation across securities and over time. Some stocks are best predicted by attention, others by sentiment, and some stocks by each at different times.

Keywords: social media, network analysis, online community, financial markets, stock prediction

A social media platform for financial discourse

In the last decade, social media analytics has become an extremely active area of research across numerous disciplines, ranging from computer science and mathematics to political science, economics, and sociology. In mathematics and computer science, researchers often study the theoretical properties of online network structures and their emergent behaviors (Erdős et al., 1960; Albert and Barabási, 2002; Newman, 2008). In the social sciences, research focuses on the *social* dimension of networks: how similar individuals are more likely to group together (e.g., *homophily*, (McPherson et al., 2001)), how online networks influence offline behavior (Victor et al., 2017; Allcott et al., 2024), and how social media can drive societal changes across domains as diverse as election outcomes (Grinberg et al., 2019) and mental health (Braghieri et al., 2022).

While numerous datasets exist for studying social media, data availability remains a significant limitation in exploring how individuals interact on social media and how these interactions drive real-world outcomes (Abdesslem et al., 2012). The most-studied social media platform, largely due to its previously generous data access policies for academics prior to 2023, is Twitter/X.¹ Twitter is a massive platform, with an estimated 560 million tweets per day and more than 350 million users.² However, access to its data is costly even for academics, and analyzing it systematically poses significant computational challenges. Consequently, any analysis of Twitter data necessarily focuses on a small subset, such as tweets related to COVID-

¹A Google Scholar search for “Twitter data” yielded more than 157,000 results as of this writing, including R and Python libraries designed for accessing, parsing, and analyzing Twitter data.

²<https://www.omnicoreagency.com/twitter-statistics>

19 (Lamsal, 2021) or those attributed to suspected Russian bots (Stukal et al., 2017).

Other social media platforms, including Twitter after its 2023 acquisition, are increasingly reluctant to provide data to external researchers, even under restrictive conditions (King and Persily, 2020). For example, the primary dataset used to study Facebook is the Crowdtangle dataset of shared URLs (Edelson et al., 2021), while user-level and post-level data remain strictly unavailable. Similarly, newer platforms such as Instagram, WhatsApp, and TikTok are studied only minimally. (For notable exceptions, see Yang (2021); Reece and Danforth (2017); Mars and Scott (2016); Valenzuela et al. (2021); Medina Serrano et al. (2020).)

In this paper, we introduce the first publicly available, *comprehensive* social media dataset, representing the complete history from 2008 through 2022 of a social media platform called StockTwits. StockTwits is a platform designed for discussions of investment strategies and financial securities trading. This dataset contains the entire history of more than half a billion posts by over 8 million users, totaling more than 167 gigabytes of uncompressed data. Financial social media is a domain where existing studies are severely constrained by data limitations. For example, three prominent studies in this area include Jaggi et al. (2021), which focuses on the stocks of the “major five FAANG (Facebook, Amazon, Apple, Netflix, Google)” companies; Bozanta et al. (2021), which performs sentiment analysis restricted to “five stocks, namely, Apple Inc. (AAPL), Amazon (AMZN), Boeing Co. (BA), Walt Disney Co. (DIS), and the SPDR S&P 500 ETF Trust (SPY)”; and Koukaras et al. (2022), which examines stock movements for Microsoft exclusively.

Literature Review

Over the past two decades, a body of literature has emerged that uses data from public online platforms such as Twitter and Reddit to predict stock market movement; for recent overviews, see Lin and Marques (2024); O’Leary (2015); Zhang et al. (2022); Nassirtoussi et al. (2014). Following Bollen et al. (2011), using crowd-sourced *sentiment* from Twitter and Reddit data to predict stock market has become one of the dominant approaches in this field. A long history of work in behavioral finance argues that crowd sentiment can affect financial outcomes significantly (Nofsinger, 2005); by accessing broad public sentiment from (micro-)blogs and social media researchers have found remarkable success in predict financial market movements,

often better than traditional methods based on random walk theory and the Efficient Market Hypothesis (EMH)(Fama, 1965).

This line of work has thrived in recent years as asset classes like cryptocurrency and meme stocks have emerged. Research incorporating novel text-mining tools to measure public sentiment (Yang et al., 2015; Guo and Li, 2019) and leveraging machine learning models has been shown to predict movements in assets' price, volatility, and volume (Oliveira et al., 2017). Our object of study is StockTwits, an increasingly popular data source in this field. With only minor differences in structure and form from Twitter data, studies based on StockTwits data have found it as good or better for studying such topics (Jaggi et al., 2021; Batra and Daudpota, 2018).

However, much of this literature is hampered by two critical limitations.

Firstly, most previous works on stock market-related social media data are constrained in scope: the sheer size of social media data often necessitates selecting only small subsets for analysis. For example, three prominent studies in this area include Jaggi et al. (2021), which focuses on the stocks of the *"major five FAANG (Facebook, Amazon, Apple, Netflix, Google)"* companies; Bozanta et al. (2021), which performs sentiment analysis restricted to *"five stocks, namely, Apple Inc. (AAPL), Amazon (AMZN), Boeing Co. (BA), Walt Disney Co. (DIS), and the SPDR S&P 500 ETF Trust (SPY)"*; and Koukaras et al. (2022), which examines stock movements for Microsoft exclusively. While these filters are reasonable given the massive amount of data generated on popular platforms like Twitter and Reddit, these stocks are far from a representative sample of the financial space. As we will show later in this paper, stocks are reasonably connected and clustered by investors' shared attention. So stock-by-stock prediction models, as most papers present, may be biased. With our unique comprehensive dataset, the potential issue could be addressed, on one hand, by a much longer list of focus stocks which includes not only the most popular ones, and on the other hand, by identifying clusters of stocks as measured by the behavior of StockTwits users.

Secondly, the real-time nature of our data set provides unique leverage over a key question in the study of digital behavior: why and do people post on the internet? Previous sentiment analytical studies of financial media presuppose that both positive and negative

sentiment could be effectively measured by text-mining on social media. However, selective self-deletion of content – users deleting their own posts after the fact – may violate this assumption. By collecting our data in real time, we retain data even after users may delete their own posts. We find that StockTwits users are more likely to post bullish than bearish messages about a certain stock, and that part of this phenomenon can be explained by users deleting bearish posts *ex post*. Retrospective analyses, then, would exclude bearish sentiment during a strong bull market, and in turn overestimate the relationship between social media sentiment and stock market behavior.

A History and Description of StockTwits

StockTwits was formed in 2008 by Howard Lindzon and Soren Macbeth, inspired by the potential that Twitter could have as a focal point for grassroots conversations about finance. It was originally designed using Twitter’s API, and though it has since developed its own infrastructure and exists as a standalone social media platform, it is still much smaller than the biggest social media platforms.³

StockTwits has a format and interface similar to Twitter’s, as seen in Figure 1. Users see a continuous feed of posts about financial markets written by other users and can follow them to prioritize seeing those users’ posts in their feed. Users can “like” a post, reply to it to start a thread, or reshare a post.

³As of the time of this writing, it has fewer than 100 employees, according to Pitchbook (<https://pitchbook.com/profiles/company/51165-19#overview>), compared to 5,500 for Twitter (<https://eulerpool.com/en/stock/Twitter-Stock-US90184L1026/Employees>) and more than 65,000 for Facebook (<https://www.statista.com/statistics/273563/number-of-facebook-employees/>).

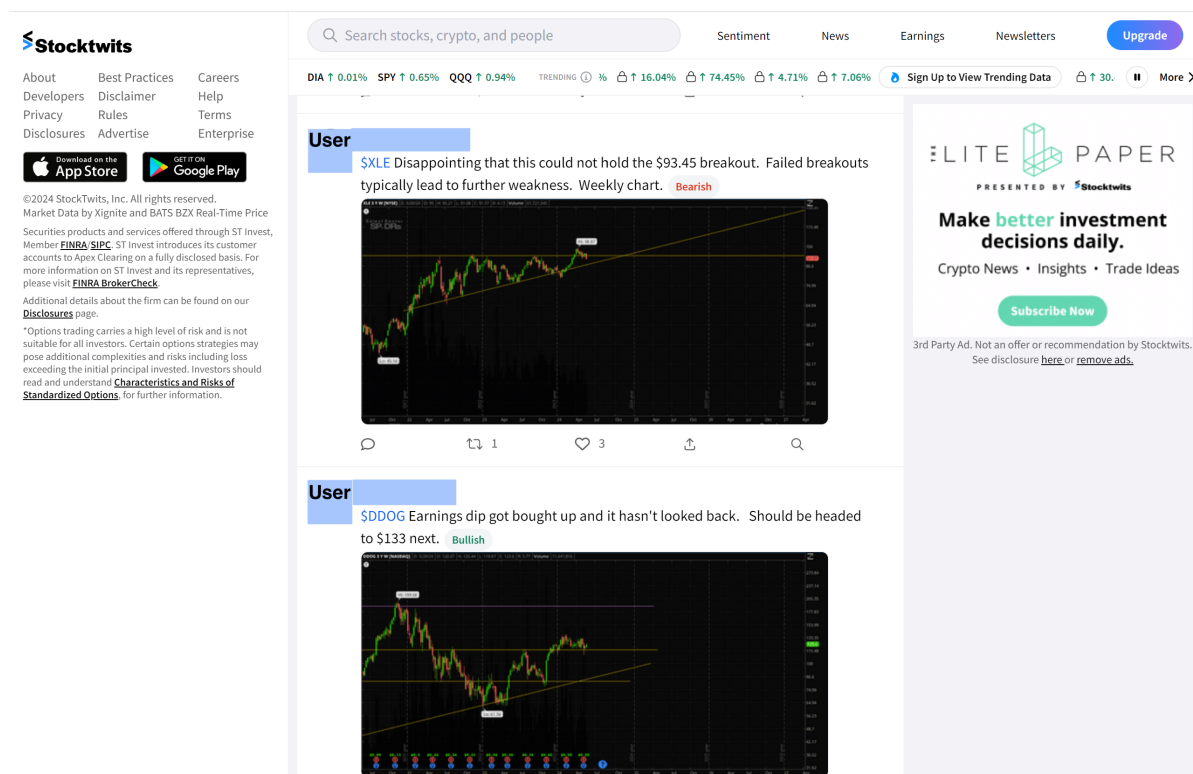


Figure 1. A screenshot of the StockTwits webpage illustrating its key features.

There are two key distinctions. First, rather than hashtags, users can include “cashtags” to indicate that their post specifically refers to a given security; users can also follow securities or cryptocurrencies, just as Twitter users can follow a hashtag. In Figure 1, we can see that user “wkalungi” includes the “\$XLE” tag in their post, indicating that they are discussing the stock chart of the Energy Select Sector SPDR Fund, an exchange-traded fund (ETF) focused on the energy sector.

Second, users can optionally tag their posts with a sentiment label: “Bullish,” indicating a positive or optimistic sentiment, or “Bearish,” indicating a negative or pessimistic sentiment. The first post in Figure 1 is labeled as “Bearish” (in red), and the second post, discussing “\$DDOG” (DataDog, a cloud-based data analytics platform), is labeled “Bullish” (in green). StockTwits collects this data and produces sentiment charts for each cashtag, which are available on the front page.

The StockTwits ecosystem also includes several other tools that are tangential to the social media platform, including a brokerage for buying securities and an exchange for trading cryptocurrencies; these data are not publicly accessible.

We find this dataset to be useful along two key dimensions. First, its comprehensiveness enables researchers to study emergent properties of large social networks that cannot be observed by looking only at a subset of the network: phenomena like virality (Han et al., 2020; Campbell et al., 2023), network contagion (Fan et al., 2020; Leister et al., 2022), and interconnectedness (Bloch et al., 2023). Second, this dataset may reveal important patterns for 21st-century financial markets and holds implications for how we understand the price discovery feature of markets (Fan et al., 2023), forecast changes in consumer or investor sentiment (Huang and Liu, 2020), and study the drivers of collective rationality or irrationality (Aloosh et al., 2021).

This paper proceeds as follows. **Data Collection and Preparation** outlines the tools we developed for collecting, cleaning, and parsing the StockTwits dataset at scale. **Descriptive Summary of the Data** introduces the dataset itself, including the key variables and some descriptive statistics about them, such as the amount of traffic over time and key topics users discuss. Our second empirical section, **Can StockTwits Users Beat the Market?**, describes an individual-level analysis of StockTwits users’ predictions and accuracy. The third empirical section, **Does StockTwits Track the Market?**, aggregates StockTwits data and compares its sentiment and attention to overall market trends. Our final empirical section, **When Does StockTwits Predict Market Movements?**, uses network analysis tools to cluster tickers, identifying subtypes of tickers for which StockTwits data predicts price. We summarize in the **Conclusion**.

Data Collection and Preparation

StockTwits has witnessed more than half a billion posts as of this writing, and while there is an application programming interface (API) to access this data, it is inefficient, and priority access is rarely granted. As such, to collect this dataset, we adopt a web scraping approach,

automatically downloading hundreds of millions of public webpages.⁴

StockTwits posts are numbered sequentially. For example, a recent post has the URL <https://StockTwits.com/MemeHub/message/575045949>, indicating that it was posted by user “MemeHub” and that the message ID is “575045949.” We collect posts sequentially, beginning with the first post and continuing indefinitely. We clean and parse each post into a tabular format, then extract or transform several key variables. These two steps are discussed in detail below in **Data Cleaning**. Note that while this paper only describes the dataset as collected through 2022, we will update the dataset available online as we collect and clean posts from more recent years.

The key challenge in collecting data from StockTwits is downloading hundreds of millions of public webpages and parsing them to retrieve data despite strict per-IP address usage limitations imposed by the web server.

We began by modifying a browser based on Firefox and recompiling it from the source code to bypass the built-in protections that would otherwise block automated data collection attempts. By customizing the browser, we ensure seamless interaction with the StockTwits website without triggering security flags.

To facilitate communication between our modified Firefox browser and the scraping infrastructure, we developed a specialized HTTP proxy (tbproxy). This proxy operates by interacting with a headless Firefox instance. By hijacking the console session of the headless browser, tbproxy redirects the session to a small web server integrated into the proxy itself. This setup is analogous to reverse-tunneling and is essential for maintaining an anonymous and secure connection (Sit, 2000). The tbproxy network infrastructure relies on an overlay network, enabling anonymous communication and the provision of unique IP addresses. This approach effectively mitigates the risk of IP-based blocking and enhances the reliability of data collection.

⁴Note that all web scraping was conducted on public sources in compliance with [University]’s office of legal counsel. All messages included in this dataset were posted by users on a social media platform and should be considered only in that context and with that intent in mind.

To streamline the deployment and management of tbproxy, we containerized it using Docker. This containerization simplifies the integration of tbproxy into our broader orchestration tool, which is based on Docker Compose. By containerizing the proxy, we ensure ease of use, scalability, and portability across different environments, which is critical for maintaining a robust and adaptable data collection system.

We developed a scrape worker responsible for the actual data extraction process. Each instance of the scrape worker accepts a batch of message IDs from StockTwits and processes them sequentially. The worker handles retries in case of failures, ensuring that transient issues do not disrupt the data collection process. Each processed batch is saved in compressed formats as JSON (‘.json.gz’) and CSV (‘.csv.gz’), offering flexibility for subsequent data analysis and integration. This compression reduces storage requirements and improves data handling efficiency.

The scrape manager plays a pivotal role in orchestrating the entire data collection system. It manages the lifecycle of the proxies, including scaling the number of running proxies up or down based on demand, monitoring their health, and assigning them to scrape workers. Each proxy is dedicated to a single scrape worker to avoid contention and ensure optimal performance. The manager also oversees the batch processing workflow, providing real-time statistics and progress updates in both JSON and HTML formats. This centralized management approach ensures efficiency, reliability, and transparency in the data collection process.

The scraping process was conducted on a high-performance virtual machine (VM) equipped with 16 cores, 64GB of memory, and 1TB of SSD vSAN storage. Each scraping run processed batches of 1000 message IDs using approximately 100 proxies. This setup ensured efficient and reliable data collection.

Data Cleaning

In total, we collected 561,219 raw CSV files, amounting to around 167GB of storage on disk. Each raw file contains 1000 sequential posts by message ID, except for messages that were deleted before our collection. Since this data is scraped, most of the fields are clean when we collect it. However, there are several key exceptions.

We perform substantial cleaning on the **symbols** variable, which contains the stock tickers tagged in the post. Upon collection, this data is stored as a list of dictionaries, and we transform it into a simple vector. For example, for a message tagging AAPL and SPY, the original entry for its **symbols** column would be:

```
[{"symbol": "AAPL", "symbol mic": "AAPL.XNAS", "title": "Apple Inc.",...}, {"symbol": "SPY", ...}]
```

We transform this complex structure into a much simpler vector of tickers:

```
["AAPL", "SPY"]
```

We perform a similar cleaning step for the **sentiment** variable, where each entry is a dictionary with only one key and the sentiment – either “Bullish” or “Bearish” – as the value. After cleaning, only the sentiment string is stored.

After these two cleaning steps, the total size of the dataset is reduced to less than 100GB. This is still prohibitive for most researchers, so we also produce several files without the full set of records. These include the following files:

1. Message ID and message text only (54GB)
2. Message ID and all variables except message text (50GB)
3. Message ID, ticker, and sentiment (5GB)

Data Records

Each record in our dataset is associated with a single StockTwits post, described in Table 1. Each post has a unique integer message ID and is associated with both the user ID of the StockTwits user who posted the message and the timestamp when it was posted. For an overview of the volume of messages over time, see Figure 4. We also provide the complete text

of the post itself. These four variables – message ID, user ID, message text, and timestamp – have no missing values.

We collect three additional variables for each post: sentiment, parent message ID, and the list of cashtags. First, as noted above, users have the option to tag each post with a sentiment label, either “Bullish” or “Bearish.” In practice, 52.8% of posts have no labels, 40.9% are labeled Bullish, and 6.2% are labeled Bearish. This partly reflects the fact that the market is generally positive during this period. Second, as with many other social media platforms, messages can be *threaded*: users can write a message in reply to a previous message, called the parent message. We include this data in the “parent_message_id” column, which specifies the Message ID of the parent message (if any). Overall, 22% of messages have no parent IDs. Finally, we extract from the message text a list of all cashtags in the message. 58.9% of messages do not include a cashtag; among the messages that do, most only include one, though some messages may include many cashtags. We detail this pattern in Figures 4 and 5.

Column	Type	Description	Note 1	Note 2
message_id	int	Unique ID for each message		
user_id	int	Unique ID for each user		
message_body	str	Message text		
date_sent	str	The date and time of the post	str in the form of year-date-datetime	The date ranges from 2008-05-27 to 2021-11-03
sentiment	str	The user-labeled sentiment of the post	NaN: 64.9%; Bullish: 28.0%; Bearish: 7.1%	
parent_mes_id	int	If the message is a reply, the message_id of the parent message	78% of messages have a parent message ID	
Symbol	list	List of “cashtags” in the message body	47.7% of messages do not tag any ticker	Most tagged messages tag only 1 ticker

Table 1: Description of Variables***Data Format and Repository***

The dataset is stored in an Amazon S3 bucket and is available for public access at <https://stocktwits-nyu.s3.us-west-2.amazonaws.com/dataset/README.md>. It is provided for convenience in a CSV file format.

The data is organized into several directories, each containing CSV files that focus on different aspects of message processing and sentiment analysis. The directory structure is as follows:

```
dataset/
  README.md
  v1
    data
      csv
        feature_wo_messages
          feature_wo_messages_000.csv
          ...
        messages
          msg_000.csv
          ...
        msg_infos
          msg_info_00.csv
          ...
        sentiments
          sentiment_00.csv
          ...
        symbols
          symbol_000.csv
          ...
        symbol_sentiments
          symbol_sentiments_00.csv
          ...
```

Each sub-directory contains specific CSV files tailored to its purpose:

- **feature_wo_messages:** Contains files like 'feature_wo_messages_000.csv', which include columns such as 'message_id', 'user_id', 'created_at', 'sentiment', 'parent_message_id', 'in_reply_to_message_id', and 'symbol_list'.
- **messages:** Contains files like 'msg_000.csv', which include columns 'message_id' and 'message_body'.

- **msg_infos:** Contains files like ‘msg_info_00.csv’, which include columns such as ‘message_id’, ‘length’, and ‘important_words’.
- **sentiments:** Contains files like ‘sentiment_00.csv’, which include columns like ‘message_id’, ‘user_id’, ‘created_at’, ‘sentiment’, and ‘symbol_list’.
- **symbols:** Contains files like ‘symbol_000.csv’, which include columns such as ‘message_id’, ‘user_id’, ‘created_at’, ‘sentiment’, ‘symbol_list’, ‘sym_number’, and ‘symbol’.
- **symbol_sentiments:** Contains files like ‘symbol_sentiments_00.csv’, which mirrors the structure of the ‘sentiments’ files.

The repository also includes a ‘README.md’ file, which provides up-to-date information on the structure, usage, and content of the data. This document is regularly updated to reflect any changes or additions to the dataset, ensuring that users have access to the most current information.

This organization facilitates efficient data retrieval and processing, allowing researchers to work with subsets of data relevant to their specific analyses without needing to load the entire dataset into memory.

There are no restrictions on its use, and it is freely available for research purposes. Instructions for accessing and downloading the data from the Amazon S3 bucket are provided on the project’s Open Data on AWS page.

Missing Data

We note in the Data Cleaning section that some messages had been deleted before we could collect them. Since message IDs on StockTwits are sequential and correspond to the date of posting, we can identify which message IDs are missing from our dataset to estimate the proportion of deleted posts over time. Figure 2 shows these proportions. As expected, older posts are more likely to be deleted: nearly 25% of all messages posted before 2010 are missing. However, from 2014 to 2020, the missing rate remained around 5%. A notable spike in the missing message rate occurs in mid-2020 and lasts until late 2021, which corresponds with the

COVID-19 surge of StockTwits users. While the exact cause of this spike is unclear, there is no evidence suggesting it is related to an uptick in bot activity. Our live data collection process began in 2023, and we expect little to no missingness going forward, as we collect posts as soon as they appear.

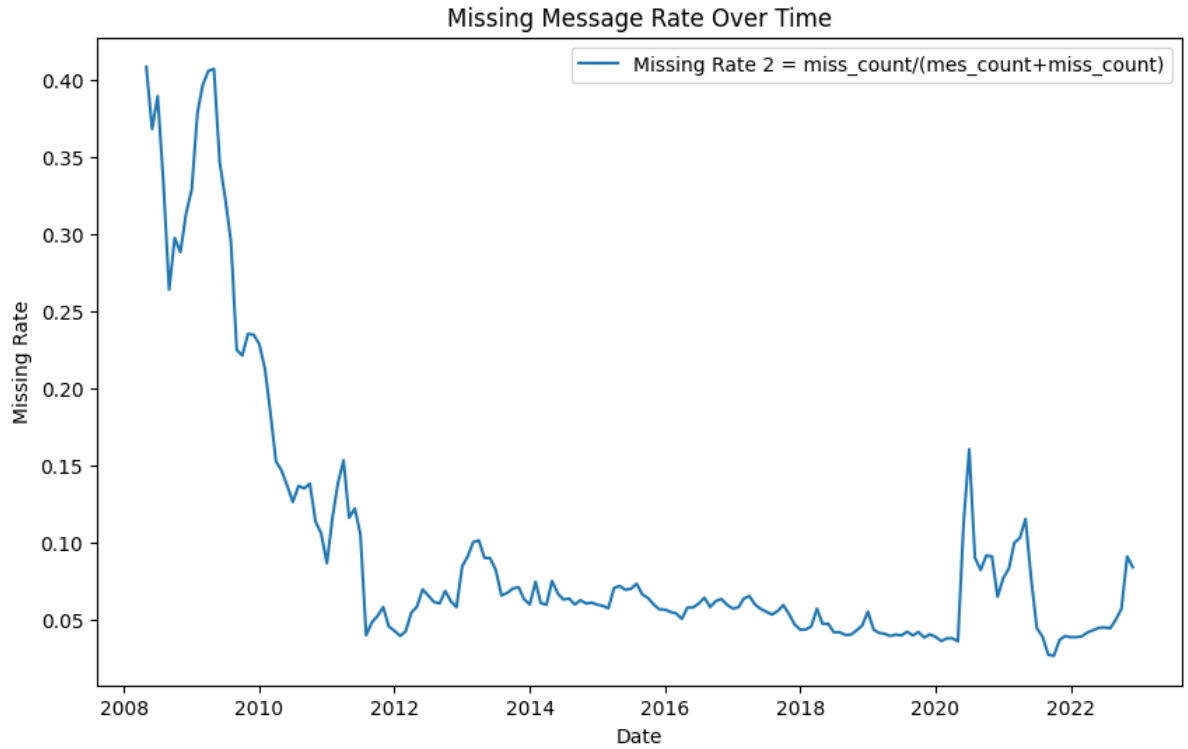


Figure 2. The proportion of deleted posts over time.

However, we might be concerned about the *sources* of missingness in this data. Posts may be missing from our data set for three reasons: StockTwits may delete posts that violate their terms of service, users may delete their own individual posts, or they may delete their entire accounts, thereby deleting all their posts. Both leave us vulnerable to forms of selection bias, for example if users selectively delete posts that, with hindsight, were inaccurate or misleading.

To study the source of missingness in our data set, we re-downloaded in 2025 a random sample of 98,588 message IDs posted between November 2021 and December 2022. Of those,

94,055 exist in our original scraped data set, while 72,786 are still available in the rescraped data set, indicating that 22.6% of posts disappeared in that period.

To start, we find that a large proportion of these missing posts come from users deleting their entire accounts, and the large increase in missing data we attribute to users who opened StockTwits accounts during the COVID-19 lockdowns and subsequent stimulus check bull market and then deleted their accounts prior to our data collection. Digging deeper, however, we do observe small differences between the content of deleted and non-deleted posts. Since the stock market performed poorly in this period, with the S&P500 going from about \$460 to about \$380 in this period, if users were deleting their incorrect posts selectively, we would expect to see that the deleted posts were more bullish than the non-deleted posts. We find the opposite. While 34.9% of the non-deleted posts are labeled as bullish and 5.5% are bearish, only 28.5% of the deleted posts are bullish and 8% are bearish, with the remaining posts untagged.

There are no statistically significant differences between the prevalence of tickers tagged in the deleted versus non-deleted posts. Furthermore, an analysis of the words that differ most between deleted and non-deleted posts is not especially informative. The words with the highest prevalence in the deleted posts relative to the non-deleted are 'ly', 'etf', 'sweep', 'members', and 'mom'; the words least popular in the deleted posts are 'calls', 'next', 'rip', 'hope', and 'could'.

Descriptive Summary of the Data

How active are users on StockTwits? The platform has experienced strong, though inconsistent, growth over its history. Figure 3 shows the monthly count of messages from the platform's inception in 2008 to December 31, 2022⁵. StockTwits grew exponentially through 2018 but then stalled at an average of around 28,000 messages daily. The right panel zooms in on the period from 2018 to the present. During the COVID-19 pandemic, when many people had more time at home and US investors received economic stimulus payments (Divakaruni and Zimmerman, 2024), usage sharply increased (red dashed line), reaching an average of over 110,000 messages per day. More recently, interest spiked in 2021 during the so-called "meme

⁵We average at the month level to smooth out the reduction in posts that occurs on weekends when typical financial markets – excluding cryptocurrency – are closed.

stock” surge, when stocks like Gamestop and AMC experienced inexplicable rallies (Costola et al., 2021), drawing attention to the role of social media in influencing financial markets. During this period, StockTwits saw nearly 300,000 messages per day on average.

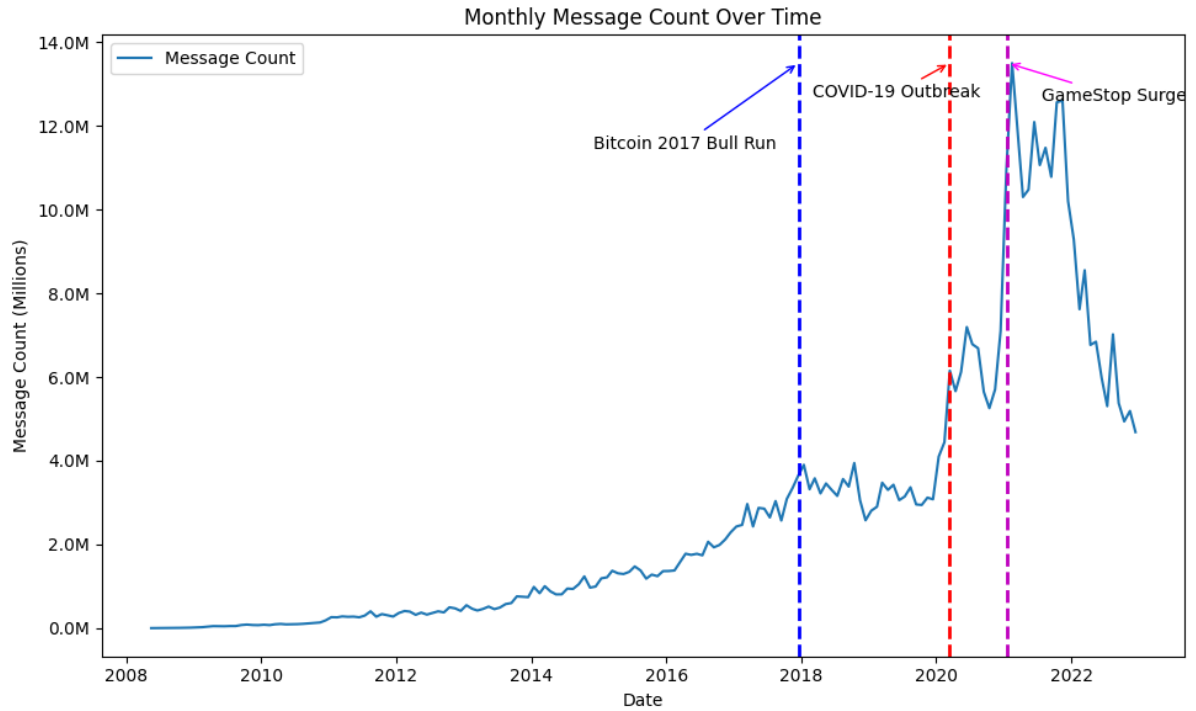


Figure 3. Monthly count of posts on StockTwits. Vertical dashed lines indicate key events related to StockTwits’ popularity.

Popular Tickers

What do users discuss on StockTwits? Our dataset includes 18,823 unique tickers mentioned as cashtags, each representing a distinct security, such as stocks, bonds, ETFs, market indices, cryptocurrencies, and derivatives. Of these, the twenty most popular tickers, which together account for almost 20% of all messages, are listed in Figure 4. The most popular include four key meme stocks — Gamestop (GME), AMC Theaters (AMC), Vinco Ventures (BBIG), and Blackberry (BB) — as well as major technology companies like Apple (AAPL), Amazon (AMZN), Meta (META), and Advanced Micro Devices (AMD), along with car manufacturers such as Tesla (TSLA), Nio Inc (NIO), Mullen Automotive (MULN), and Lucid Group (LCID).

Cryptocurrencies like Bitcoin (BTC.X), Dogecoin (DOGE.X), and Shiba Inu Coin (SHIB.X) also feature prominently. The remaining tickers are a mix of a biotech company (OCGN), an energy company (CEI), a cannabis company (SNDL), and two indices: SPY, the S&P 500 stock index, and ES-F, an S&P 500 futures fund.

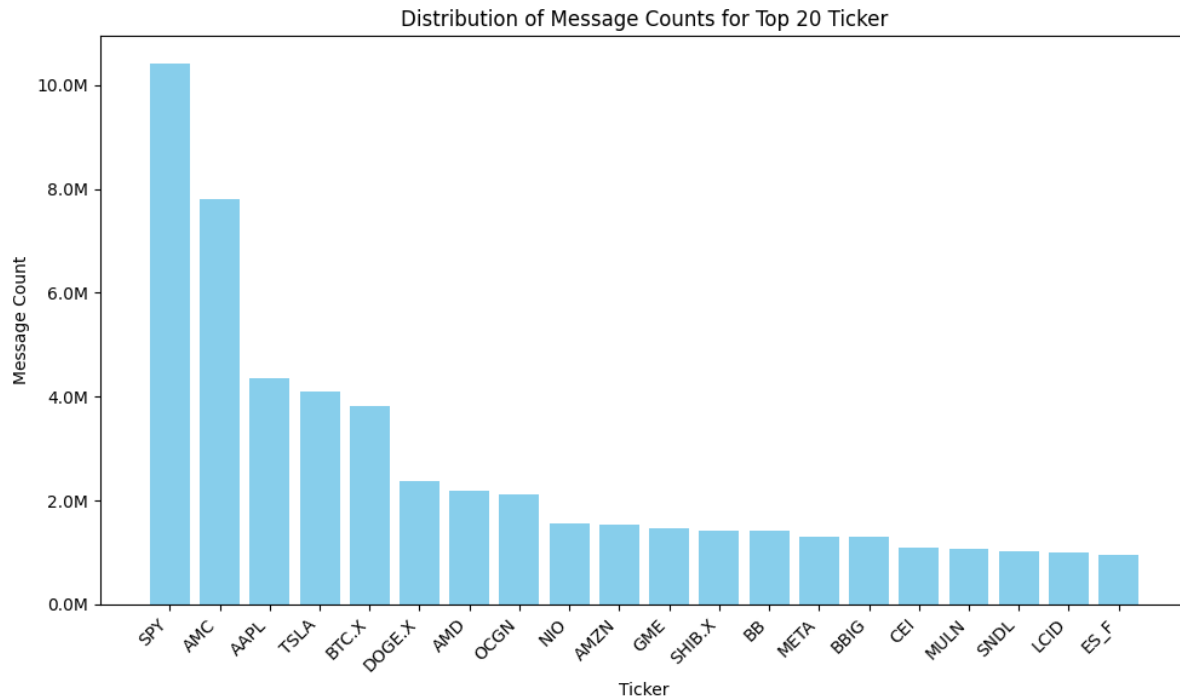


Figure 4. Some tickers are more popular than others on StockTwits; the top 20 most popular tickers each represent a large proportion of overall messages.

The attention paid to these twenty most popular tickers varies over time, as shown in Figure 5. Each bar represents the number of months a ticker appeared in the monthly Top 20 list, which is a measure of consistent popularity. We observe that SPY is both the most-discussed ticker overall and has appeared in the Top 20 the most frequently, nearly 175 months. The second-most popular ticker overall is AMC, but it ranks 24th for months in the Top 20, indicating its popularity is more sporadic. The blue bars represent tickers that are also in the Top 20 overall, while the grey bars represent those that are not.

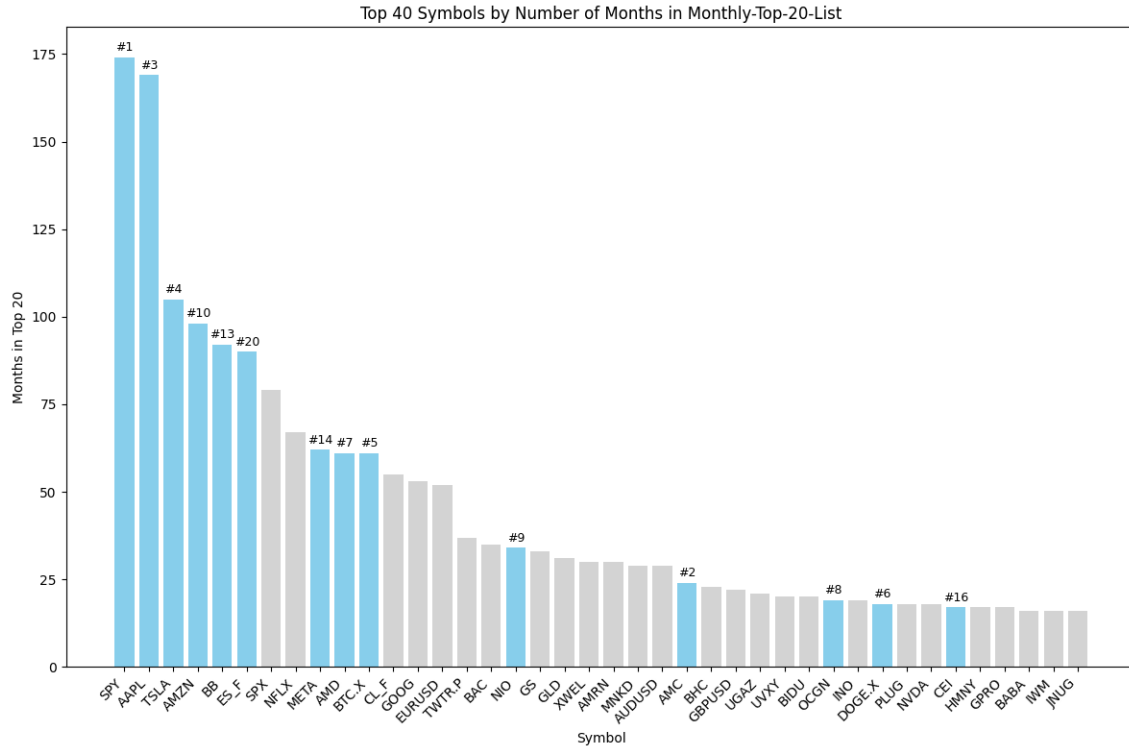


Figure 5. Each bar indicates the number of months that a ticker was in the monthly Top-20 Tickers list. Numbers above each bar indicate the overall rank of that ticker in total message count.

While AMC and GME are most discussed during the meme stock rallies, tickers like Apple, Tesla, and SPY maintain a much steadier presence.

Sentiment-Specific Lexical Analysis

To identify lexical features that are most indicative of bullish and bearish sentiment in StockTwits posts, we employed a supervised text classification approach. With the sentiment-labeled data for stock AMC from 2021-05 to 2021-07 (a bullish period) and from 2021-11 to 2022-01 (a bearish period). Each message in the dataset is associated with a user-labeled sentiment indicator, where 1 denotes bullish sentiment and -1 denotes bearish sentiment. Our objective was to extract and rank the most discriminative words and phrases (n-grams) that predict the

sentiment label.

Preprocessing. We first applied standard text preprocessing procedures, including lower-casing and tokenization. Common English stopwords were removed, except for finance-specific vocabulary (e.g., “buy”, “sell”, “short”, “long”). Both unigrams and bigrams were considered in order to capture sentiment-relevant multi-word expressions (e.g., “price target”, “short squeeze”).

Feature Representation. The processed messages were transformed into a numerical representation using the Term Frequency–Inverse Document Frequency (TF–IDF) scheme:

$$\text{tfidf}(t, d) = \text{tf}(t, d) \times \log \left(\frac{N}{\text{df}(t)} \right),$$

where $\text{tf}(t, d)$ is the term frequency of term t in document d , $\text{df}(t)$ is the number of documents containing t , and N is the total number of documents. This weighting emphasizes terms that are frequent in a given class while reducing the influence of common terms across classes.

Modeling and Interpretation. We fit a logistic regression classifier with ℓ_2 regularization to the TF–IDF vectors, treating bullish sentiment as the positive class and bearish sentiment as the negative class. The model learns a coefficient β_j for each term j ; the sign and magnitude of β_j indicate the direction and strength of association between the term and the sentiment label:

$$\hat{y} = \sigma \left(\beta_0 + \sum_{j=1}^p \beta_j x_j \right),$$

where $\sigma(\cdot)$ is the logistic function and x_j is the TF–IDF weight of term j .

Term Ranking. We ranked all by their corresponding logistic regression coefficients. Positive coefficients indicate a stronger association with bullish sentiment, whereas negative coefficients indicate an association with bearish sentiment. The top-ranked terms provide interpretable lexical cues that distinguish the two sentiment classes.

We present these results in Table 2. The top section shows the unigrams most predictive of bullish and bearish sentiment for AMC Theaters (ticker AMC), a key meme stock, during its rapid rise and fall. Bullish terms include “bear trap”, a term for a deceptive and false technical indicator that a stock’s price will decrease; “hodl” and “lfg”, meme stock mantras; and “shitadel”, a reference to the hedge fund Citadel known for shorting meme stocks. Bearish terms include “baggies”/“bagholders”, terms for those who lose money when a stock crashes; and generally pessimistic terms like “delusion”, “overvalued”, and “junk.”

The bottom half of Table 2 shows the top bullish and bearish terms for related the S&P 500 ETF ticker SPY during the second half of 2021. Bullish terms included “calls 09” and “calls 12”, indicating call options; “bear trap”; and a number of other terms derogatory toward pessimistics like “dumb bears” and “greedy bears”. Likewise, the most bearish words included “puts 09”/“puts 08”, a reference to put options; and terms derogatory toward bulls like “spy bulltards”, “perma bulls”, “sorry bulls”.

Table 2: Top discriminative unigrams (AMC, 2021-05 to 2022-01) and bi-grams (SPY, 2021-07 to 2022-01) for bullish and bearish sentiment. Positive coefficients indicate bullish association, negative coefficients indicate bearish association. Only top 15 terms per category are shown.

AMC (Unigrams) Bullish		AMC (Unigrams) Bearish	
Term	Coef.	Term	Coef.
fuckery	8.762	baggies	-13.599
bear trap	8.036	bagholders	-12.169
shitadel	8.012	pumpers	-11.254
fud	7.953	moazzz	-10.628
relax	7.417	bearish	-9.746
scare	6.790	monkeys	-9.337
hold	6.769	junk	-9.281
fake bulls	6.741	bbbb_dddd	-9.164
fucking	6.212	pumper	-9.101

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AMC (Unigrams) Bullish		AMC (Unigrams) Bearish	
Term	Coef.	Term	Coef.
patience	6.044	bulls	-8.711
hodl	5.873	100k moass	-8.301
new apes	5.853	delusion	-8.097
lfg	5.748	overvalued	-7.860
bears	5.743	sheep	-7.809
holding	5.690	delusional	-7.740

SPY (Bigrams) Bullish		SPY (Bigrams) Bearish	
Term	Coef.	Term	Coef.
discord io	9.571	puts 09	-8.195
calls 09	8.849	bull trap	-5.598
calls 12	5.588	spy bulltards	-4.993
mrna bntx	5.502	bulls think	-4.921
low float	5.438	puts 09 17	-4.920
aabb spy	5.258	bulls saying	-4.633
calls 09 17	4.954	puts 08	-4.621
bear trap	4.739	non mkt	-4.602
whew spy	4.712	444 puts	-4.437
dumb bears	4.696	mrna https	-4.393
bye bears	4.571	bear flag	-4.284
sorry bears	4.427	perma bulls	-4.181
bears say	4.410	tsla mrna	-4.176
bears think	4.344	sell signal	-4.161
bears getting	4.329	puts printing	-4.150

An interesting implication of these terms is the formation of bullishness and bearishness,

both overall and with regards to specific stocks, as identity categories akin to partisanship or sports teams. We hope researchers will explore this phenomenon in future work.

Can StockTwits Users Beat the Market?

In **Data Records**, we described the cashtags that users can include to reference specific securities as well as the sentiment tags that users can include to label their messages as bullish or bearish. Taking these labels as predictions about the referenced securities, how well do StockTwits users predict stock price movement? This analysis builds on a burgeoning literature suggesting that StockTwits can be a powerful predictor of financial movement in the aggregate (Al Nasser et al., 2014; Nouri and Tjortjies, 2021; Jaggi et al., 2021; Batra and Daudpota, 2018; Fallahgoul, 2020; Gupta and Chen, 2020). However, while these papers use limited subsets of StockTwits data, we bring the entire data set to bear on this question.

For each message with a cashtag⁶ and a sentiment label, we merge in financial time series from the Yahoo Finance API via the `yfinance` Python library. Using the timestamp variable, we identify the security’s price when the user posted the message. Next, we calculate whether the security had increased or decreased in price 1, 3, 7, 14, 28, and 30 days later. Finally, we aggregate to the user level. Each user receives accuracy scores for each time delay indicating what percentage of their messages accurately predict stock movement within the given period.

For example, if a user posts a bearish (pessimistic) message about a given stock but that stock goes up for the next three days, that would count *against* her predictive success at $T + 1$ and $T + 3$; if the stock plummets a few days later and does not recover, that would count *toward* her accuracy at $T + 7$, $T + 14$, and $T + 30$. We then aggregate these accuracy statistics across all of a user’s posts to estimate how often their predictions are correct after 1 day, 3 days, and so on.

Figure 6 presents density plots for user accuracy at each timescale. The dotted vertical line at 0.5 indicates coin-flip accuracy: if a user randomly guessed which stocks would go up and down, their accuracy at all timescales would be 0.5⁷. While most users’ accuracies are

⁶For this analysis we only consider cashtags referencing one of the 50 most-tagged securities.

⁷Over this period the market was largely positive, so in practice guessing that every security will

clustered around 0.5 or below, a handful of users perform much better than guessing; quite a few users perform much *worse* than guessing. The thinnest tails are for the $T + 1$ distribution, while the fattest tails are for the $T + 30$ distribution. This makes intuitive sense: stock prices are quite variable in the short term but in the long term reflect underlying fundamentals. Note, as well, that during this period the market was largely positive, so a user who always guessed that a stock would go up would be correct far more than 50% of the time.

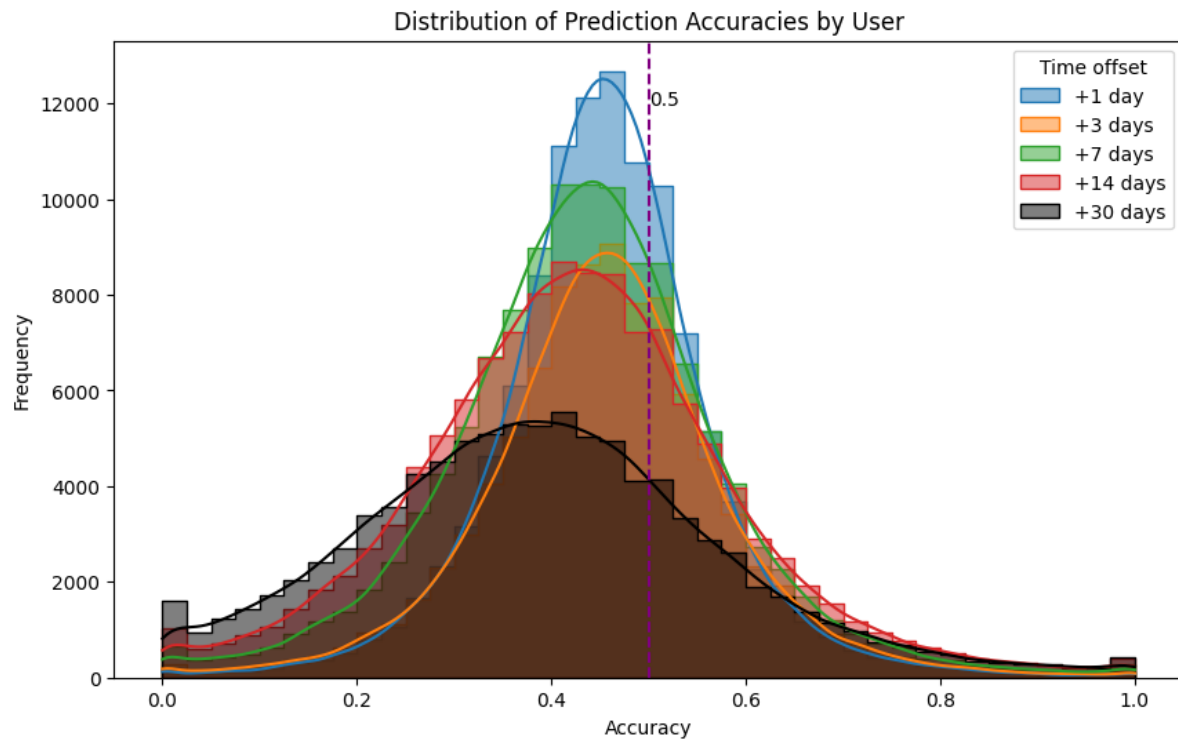


Figure 6. Density plot of users' predictive accuracy at different timescales. Accuracy one day in advance (in blue) is more concentrated around 0.5 (vertical line) than accuracy one month in advance (dark brown).

This finding that users systematically underperform the market is paradoxical. It is very difficult to find consistent signals that outperform the market. This suggests that it should be difficult to find consistent signals that underperform the market as well, since the always increase would have a much higher accuracy than 50%.

inverse of that signal would outperform the market⁸. Any individual trader who, over the long run, underperforms the market should be highly valuable.

To verify that these users are systematically under-performing or over-performing the market, rather than getting unusually lucky or unlucky, we present an analysis in Table 4. We separately calculate each user’s accuracy in the first half of their posts and in the second half, across different time horizons. If traders who perform better than guessing are simply getting lucky, we would expect that their performance during the first half would be uncorrelated with their performance in the second half. The first row shows that among users who perform worse than guessing over a 1-day time horizon in the first half of their posts, 71.58% also perform worse than guessing in the second half of their posts. In the second row, we see that among users who performed better than guessing in the first half, 61.68% also perform better than guessing in the second half. While there is substantial reversion to the mean in users’ predictive performance, many users systematically overperform, and even more systematically underperform.

Does StockTwits Track the Market?

While the median user may not be able to predict stock price movement better than guessing, or may in fact underperform it, it remains possible that users in the aggregate might still do so (Jame et al., 2016). We explore that possibility next, comparing both the *count* of messages referencing a security and the proportion of those messages labeled as “bullish.”

The first set of results is shown in Figure 7. Across ten plots, one for each of the top ten cashtags mentioned in the **Popular Tickers** section, we plot the security’s price (in blue) against the count of StockTwits posts including that cashtag (in red).

In some cases, as in Bitcoin (BTC.X; first column, third row), the price and the number of messages track quite closely. For others, like AMD (first column, fourth row), the relationship is much weaker. Still others track closely for some periods and less so for others, as with NIO (bottom left).

⁸For a discussion of this phenomenon, see <https://xkcd.com/2270/>.

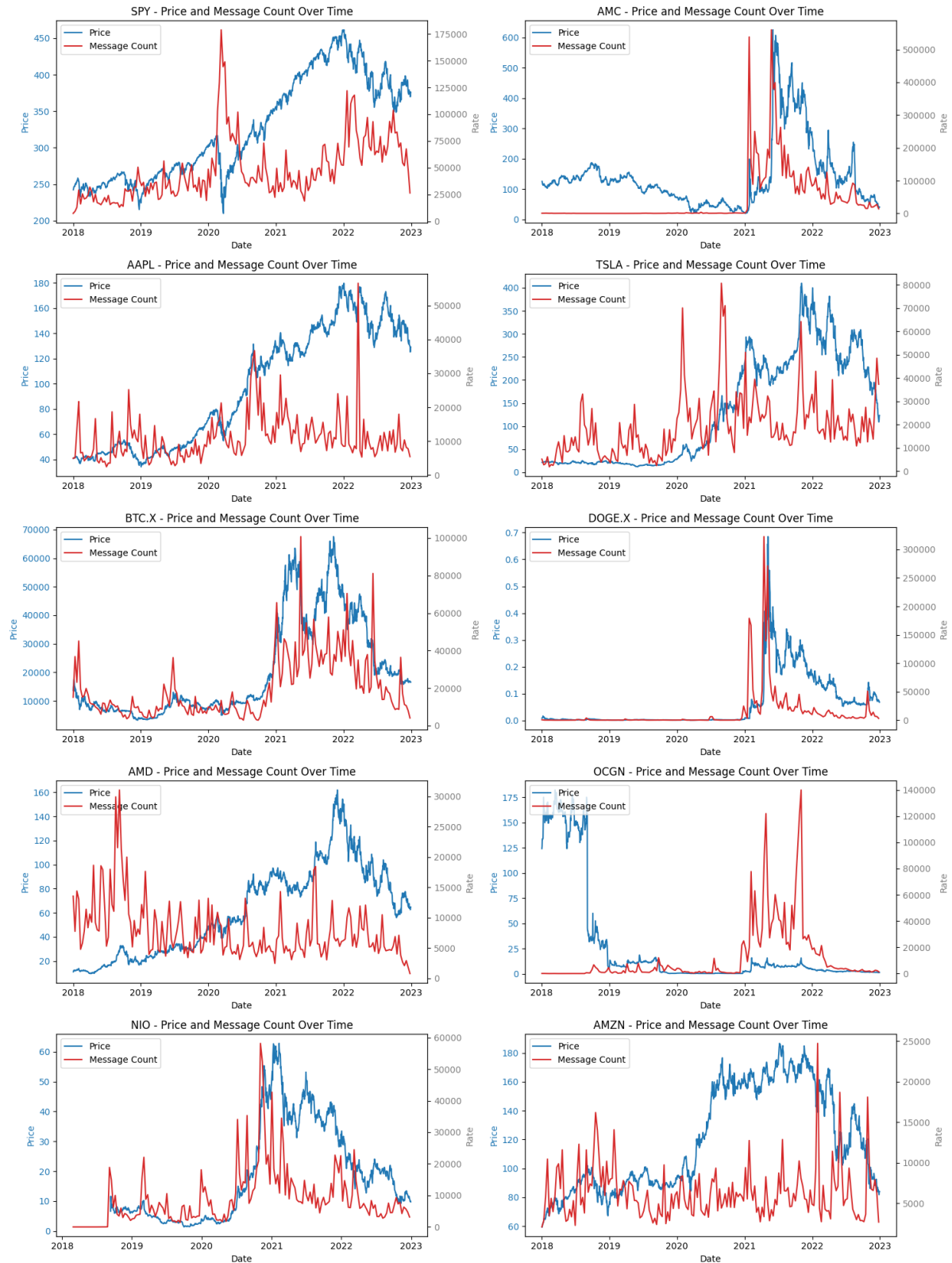


Figure 7. Message Count (10-day total) versus real stock price for a sample of popular tickers.

Time Offset		Second ≤ 0.5	Second > 0.5
1 day	First ≤ 0.5	71.58%	28.42%
	First > 0.5	61.68%	38.32%
3 days	First ≤ 0.5	71.39%	28.61%
	First > 0.5	63.53%	36.47%
1 week	First ≤ 0.5	75.65%	24.35%
	First > 0.5	64.69%	35.31%
2 weeks	First ≤ 0.5	76.45%	23.55%
	First > 0.5	65.09%	34.91%
1 month	First ≤ 0.5	80.37%	19.63%
	First > 0.5	66.91%	33.09%

Table 4: Percentage distribution of second correctness conditional on first correctness threshold, by time offset

Next, we compare prices to the daily proportion of “bullish” messages for each security, as the proportion of bullish messages with that cashtag divided by the total number of messages with that cashtag. We show ten such plots in Figure 8, one again for each of the top ten cashtags mentioned in **Popular Tickers**. Each security’s stock price is shown in blue, while its bullish rate is in green.

We observe a lot of variation across securities. In general, the correlation becomes closer over time as the number of active StockTwits users increases. In Bitcoin (second row, second column), for example, the correlation is very weak until 2021, when it suddenly becomes very strong. SPY, on the other hand, continues to correlate weakly for the complete time series.

Table 5 summarizes the above results and presents the correlation rate between a security’s price change and each of two StockTwits-derived measures: the Bullish rate (middle column) and the message count (right column). We observe that price change and Bullish rate are always positively correlated, and often very strongly: the correlation is never lower

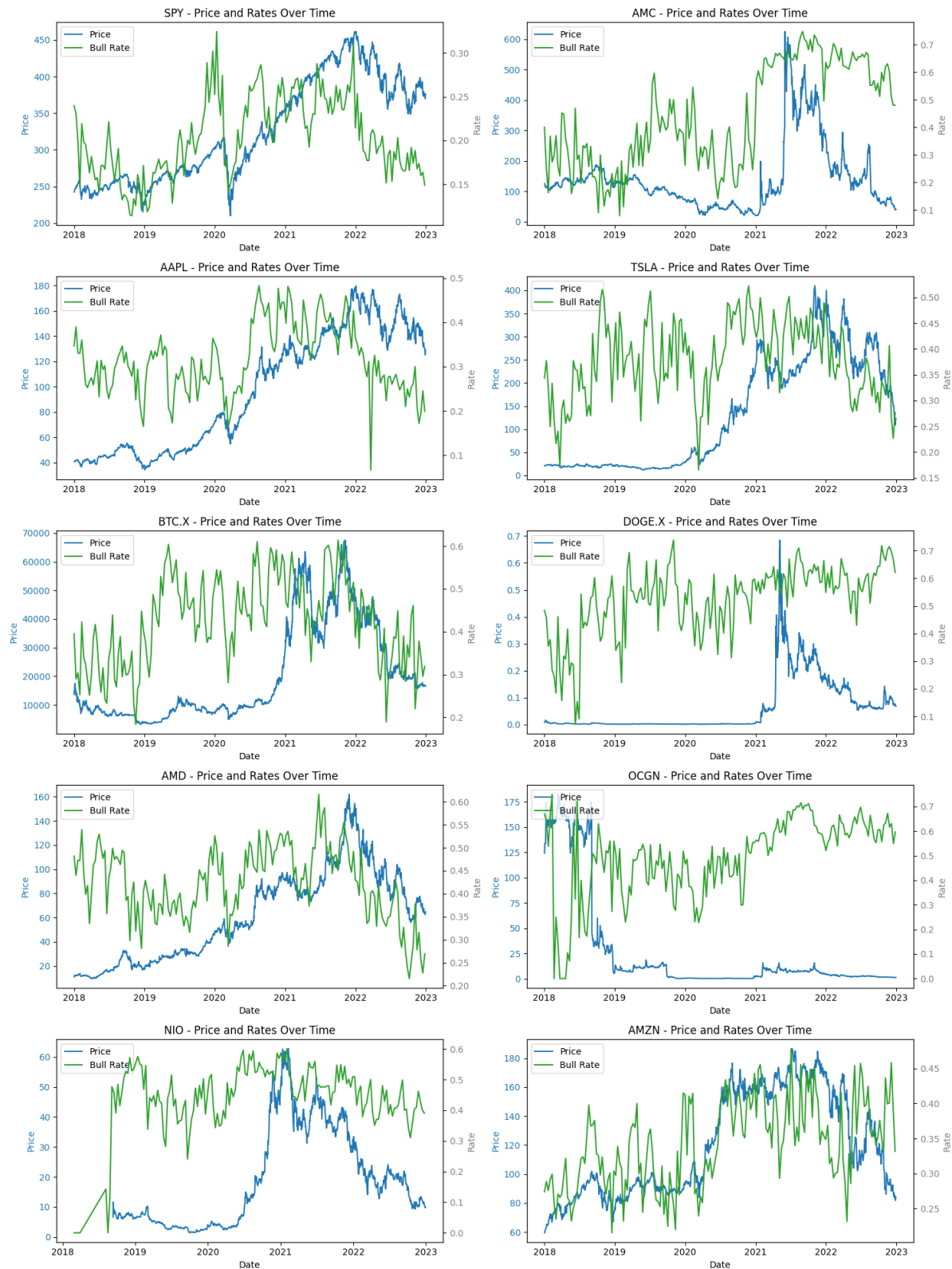


Figure 8. Bullish rate (10-day average) versus real stock price for a sample of popular tickers.

than 0.12 (for DOGE.X, the cryptocurrency Dogecoin) and reaches as high as 0.57 (BTC.X, the cryptocurrency Bitcoin). The correlation between price change and message count is more variable: as low as -0.34 for Facebook owner Meta and as high as 0.69 for Dogecoin.

Table 5: Correlation for Selected Symbols

Symbol	Price Change vs Bullish Rate	Price Change vs Message Count
SPY	0.24	-0.17
AAPL	0.25	-0.02
BTC.X	0.57	-0.17
AMD	0.39	0.18
AMZN	0.21	-0.08
BB	0.32	0.20
BBIG	0.15	0.14
SNDL	0.31	0.46
NFLX	0.16	-0.24
BABA	0.45	-0.09
XELA	0.17	0.20
CENN	0.17	0.37
INO	0.30	0.32
AMC	0.15	0.42
TSLA	0.47	0.03
DOGE.X	0.12	0.56
NIO	0.51	0.17
GME	0.14	0.69
META	0.40	-0.34
MULN	0.45	0.57
LCID	0.39	0.36
IBIO	0.18	0.39
XWEL	0.16	0.38
WKHS	0.21	0.22
NVDA	0.27	-0.03

In sum, we find significant evidence that aggregate StockTwits data can track securities price changes; we dig deeper into whether they can also *predict* securities prices next.

When Does StockTwits Predict Market Movements?

There is much variation across stocks and cryptocurrencies, and across time, in how well StockTwits aggregate data correlates with price movement. In this section, we unpack that variation to study whether StockTwits users in the aggregate can be useful for price forecasting, and if so, for which stocks. We do this in three steps. First, we compute a *network* of stock tickers, linked by users' attentions, and provide a web-based visualization tool for that network. Second, we identify clusters of stocks within those networks and calculate each stock's centrality. Third, we conduct Granger causality tests to determine whether there are subtypes of securities (and, by implication, users) for which StockTwits sentiment and attention predict price movements. Note that while Granger causality tests have the word "causality" in their name, we do not use them causally in this context, and a robust literature generally objects to their use in causal inference (see e.g. Friston et al. (2014)). Instead, we use these tests to identify which of two time series appears to predict the other.

This analysis begins with two observations. First, investors pay attention to (and post about) multiple stocks, often thematically linked. Second, different stocks attract different types of investors. An investor who works in the technology sector may primarily invest in small-cap tech firms, using her specialized expertise to make informed bets; other investors may primarily invest in meme stocks like Gamestop, basing their investment decisions on Hayek's "animal spirits". In general, we might expect small, niche stocks to have fewer but more specialized investors, and that those investors' posts will hold more information about the market. We might, in contrast, expect StockTwits conversations around large meme stocks, whose movements are driven by market sentiment, to predict prices better. At a minimum, these hypotheses suggest that the network structure of StockTwits holds important implications for stock price prediction. We begin to adjudicate these conflicting hypotheses here.

For this analysis, we construct a dynamic network using 100 weeks of StockTwits data from March 2018 to February 2020. This was a particularly stable time for StockTwits with around 700 popular tickers, before COVID and the birth of meme stocks. Within each weekly

snapshot, we construct a network with nodes as tickers. Nodes with stronger edges between them are those with more users who mention both in a given week; users who mention many different tickers are down-weighted.

Specifically, the edge weight between stocks i and j is given by Equation 1:

$$\text{EdgeWeight}_{ij} = \sum_{u \in U_{ij}} \frac{2 * \text{Min}(\text{Message}_{ui}, \text{Message}_{uj})}{\text{TotalMessage}_u} \quad (1)$$

To give some intuition for this measure, consider a user where 20% of their posts are about BTC, 30% are about AAPL, and 50% about all other stocks. This user contributes $2 \times \text{Min}(0.2, 0.3) = 0.4$ units of weight to the edge between BTC and AAPL; the overall edge weight is the sum of this value across all users. This method ensures that each user counts as, at most, 1 unit of weight, meaning it is not susceptible to bot or spam accounts that may mention many tickers or a single ticker many times; all users are weighted equally. It also avoids bias from disproportionate attention: if a user posts 10% about AAPL and 90% about BTC, only $0.1 \times 2 = 0.2$ units of weight is added to the edge. And it provides a convenient and interpretable way to cull weak edges for visualization, as seen in Figure 9.

This figure is from a snapshot from the 60th week in our sample, early summer of 2019. Inside the red inset, we see two main clusters. The bottom right is a cluster of primarily cryptocurrencies: Ethereum (ETH.X), Litecoin (LTC.X), and Bitcoin (BTC.X), as well as MARA Holdings Inc (MARA), a digital assets firm. The cluster to the top left has primarily large-cap technology companies: Telsa (TSLA), Meta Inc (META), Amazon (AMZN), Nvidia (NVDA), Uber (UBER), and others. Nodes here are colored by the “bullish rate”, where lighter nodes indicate more optimistic sentiment on StockTwits. The cryptocurrency cluster is significantly more bullish than the technology cluster.

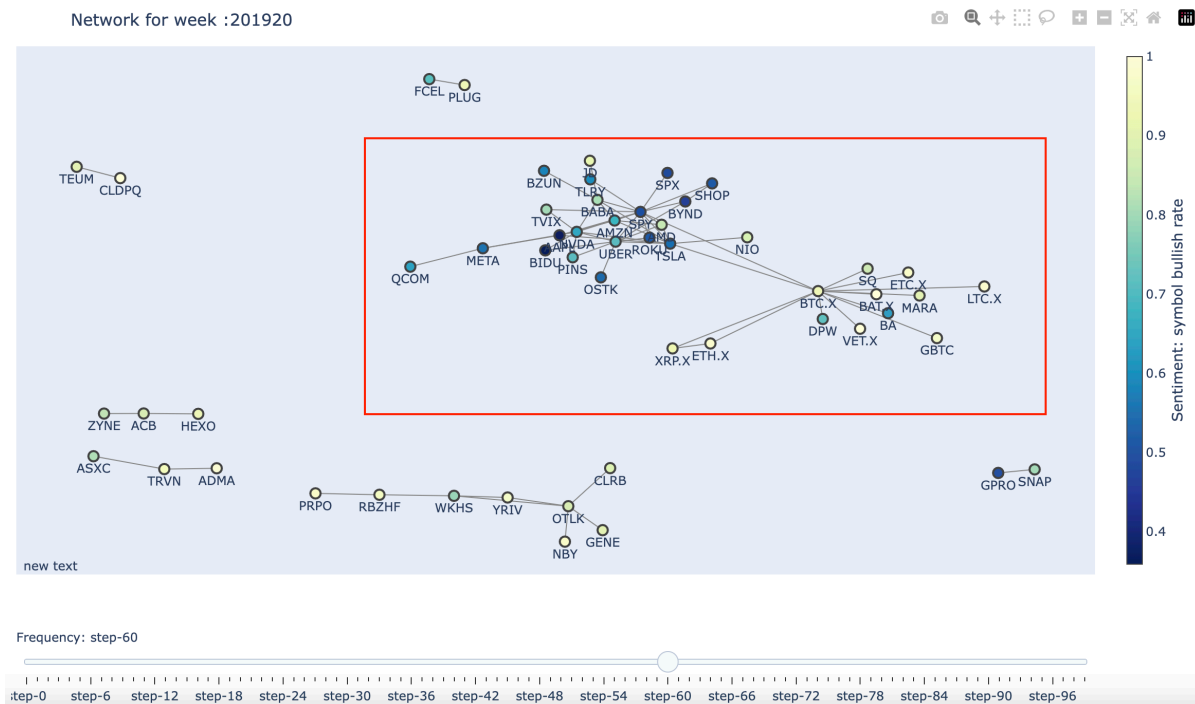


Figure 9. A Sample of Stock-stock Dynamic Network Made for this Study

Next, we compute from the network measures of each ticker's centrality – betweenness and degree – overall and for each weekly snapshot. If a ticker has high degree centrality, that ticker occupies a central position in the network: many investors pay attention to this ticker regardless of which other securities they track. SPY, QQQ, and other broad index funds are in this category. On the other hand, if a ticker has high betweenness centrality, it means the ticker occupies a bridge position in the network: this ticker is tracked by two or more different groups of investors. For example, BTC is followed closely by investors who focus on big tech companies as well as investors who focus mainly on cryptocurrencies. Note that these measures are not mutually exclusive: BTC and SPY have high values of both.

Granger testing

While network science is rich in theories articulating the effects of network position (Lewis, 2011), our analysis focuses on the narrow question of whether the network structure is an

effective predictor for individual ticker's price movement. We use the Granger causality test as a measurement device to test whether StockTwits activity is predictive of stock price among more central nodes (Granger, 1969; Shojaie and Fox, 2022).

The Granger causality test (which, again, we do not use for causal inference) evaluates whether a time series X_t , in our case StockTwits sentiment or attention, provides statistically significant information for predicting another series Y_t , in our case stock price. It compares an autoregressive model for Y_t (Equation 2) to an augmented model that includes X_t (Equation 3).

The autoregressive model takes the form

$$Y_t = \alpha_0 + \sum_{i=1}^p \alpha_i Y_{t-i} + \epsilon_t \quad (2)$$

where p is the number of lags, α_i are coefficients, and ϵ_t is the error term.

The augmented model with X_t includes lagged values of X_t as additional predictors:

$$Y_t = \beta_0 + \sum_{i=1}^p \beta_i Y_{t-i} + \sum_{j=1}^q \gamma_j X_{t-j} + \eta_t \quad (3)$$

where q is the number of lags for X_t , and γ_j are the coefficients for X_t 's lagged terms. Under this setup, the null hypothesis of:

$$H_0 : \gamma_1 = \gamma_2 = \dots = \gamma_q = 0$$

would suggest that X_t does not Granger-cause Y_t . The alternative hypothesis, H_1 , suggests at least one $\gamma_j \neq 0$, implying X_t improves prediction. This hypothesis test uses the F -statistic:

$$F = \frac{(RSS_r - RSS_u)/q}{RSS_u/(T - p - q - 1)}$$

Where RSS_r and RSS_u are the residual sum of squares for the restricted and unrestricted models, respectively, T is the number of observations, and p and q are the number of

lags for Y_t and X_t . If F exceeds the critical value, H_0 is rejected, indicating that X_t Granger-causes Y_t .

Granger Testing and Network Position

We study 630 of the most popular stocks and cryptocurrencies to test whether attention or sentiment predict price movements, and whether degree centrality or betweenness centrality moderate that relationship, 3150 total statistical tests (with time-lag of 1 to 5 days). Taking 3-day time-lag as an example, we show these results of these tests in Figures 10 and 11. The first panel in Figure 10 shows the proportion of tickers for which the Granger causality test suggests that stock attention predicts price movement. Overall, the proportion is 30%, given by the dashed horizontal line. But there is an interesting variation by degree centrality: for securities in the middle quartiles of degree centrality, attention predicts price movement much more consistently than the most or least central quartiles. However, we observe a different pattern on the right panel of that figure. Attention predicts price movement best among the tickers with the highest betweenness centrality.

Compare these results to Figure 11, which displays the results of Granger causality tests of sentiment, rather than attention, on price movement. Far fewer of these tests are statistically significant, especially considering that we expect 5% of the tests to be significant by random chance. There is weak evidence that among tickers with a lower degree centrality sentiment more often predicts price, and that among tickers with high betweenness centrality sentiment more often predicts price. However, the magnitudes are much lower than for the attention measures.

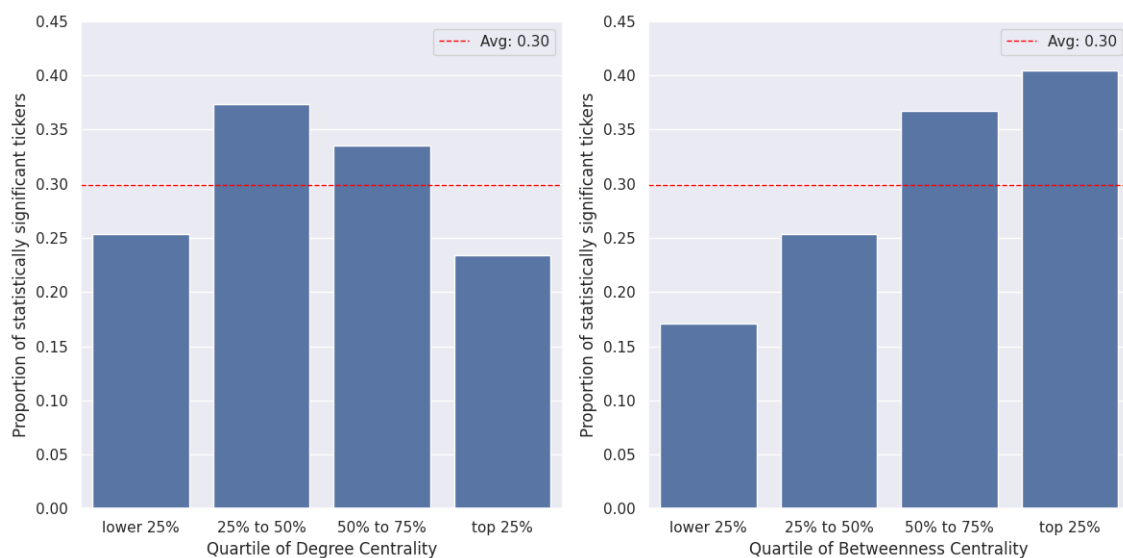


Figure 10. Attention on StockTwits predicts price movement more consistently for tickers with median degree centrality and high betweenness centrality.

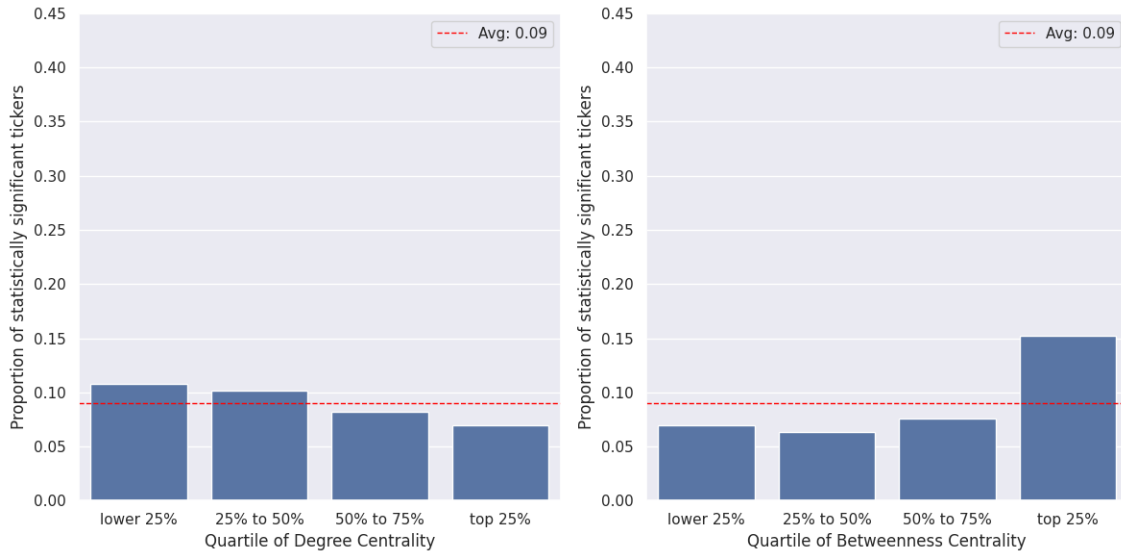


Figure 11. Sentiment on StockTwits predicts price movement more consistently for tickers with low degree centrality and high betweenness centrality, but overall much less than attention does.

Conclusion

This article is the first to systematically describe the dynamics of financial behavior on social media. Across three analyses, we find that individuals and aggregations of users can meaningfully predict stock prices, unpacking previously unconsidered heterogeneity. We also introduce a large, publicly available social media data set in a clean, tabular format suitable for any data analysis software, with additional variables that allow for network analysis and natural language processing. Uncompressed, the data set is 167 gigabytes in size; it is just over 50 gigabytes in CSV format.

We argue that this data set holds promise for diverse applications: stock forecasting, community detection, and sentiment analysis (Bozanta et al., 2021). In some cases, this data allows research that builds on existing literatures, enhancing them with richer data; in other cases, it enables entirely new research questions and analyses. The network structure of the data, in particular, opens valuable research opportunities about financial infor-

mation contagion, virality, and even fraud detection. For example, how does network structure relate to predictive accuracy? How does bullish or bearish sentiment propagate through the network? Can we identify fraudulent strategies, such as pump-and-dump schemes, using this data? We have provided analysis code that researchers can use as a starting point to extend our analysis for their own purposes. This code is available on GitHub: <https://github.com/jaxingjili/StockTwits-Data-Processing-and-Analysis/>.

However, researchers should cautiously note several potential drawbacks of this data set in conducting their own research. These drawbacks relate to sentiment labels, missing data, and strategic behavior.

As noted above, users select whether and how to label their own posts' sentiment. As such, there may be selection bias in *whether* a label is applied, and posts may be labeled incorrectly either by mistake or for insincere or strategic reasons. This behavior may harm the accuracy of sentiment models produced using this data set. Secondly, data may be missing for a number of reasons. Users may delete their own posts, and posts may get deleted when a user's account is deleted, and it is not always possible to distinguish between the two. In Figure 2, we see a steady downward trend in missing data followed by a spike in mid-2020 lasting approximately 14 months; it is not clear what caused that increase in missing data. If it was caused by astroturfed bot accounts created to fuel interest in specific stocks, and those accounts were subsequently deleted, this behavior may also impact the ability to draw inferences from this data set.

Finally, users may behave strategically in what and how they post on StockTwits. Aside from bot accounts, StockTwits users may have incentives to misrepresent their beliefs. A user might purchase a stock and then post about it, increasing interest in it and driving up its price, colloquially known as a "pump and dump". A few prominent StockTwits users with large followings have been known to materially influence stock prices with their posts.

Code availability

The access to the dataset, and all code used for analyses are available on GitHub without restriction at <https://github.com/Jaxingjili/StockTwits-Data-Processing-and-Analysis>. Analyses were performed mainly in Python.

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